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隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

Digital economy is measured through several records: the transaction value platforms process, the revenue they recognize, national e-commerce statistics and payment data. These records are often used interchangeably, yet each captures a different part of the same activity. This study asks how much economic activity remains invisible when digital platforms are measured through their reported revenue. I define the invisible wedge as the difference between transaction value and platform-recognized revenue and follow it across Indonesian platforms from 2020 to 2025. In 2023, Tokopedia, Shopee and Grab processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue. The relationship is not stable: among the platforms that report both figures themselves, revenue grew faster than transaction value in six of eight year-to-year comparisons and moved against it in two. National statistics and payment records disagree with the platform picture and with each other. BPS-Statistics Indonesia reports a marketplace channel smaller than two platforms' own transaction value and places 98.5 percent of 2023–2024 e-commerce growth outside it, while payment measures grew up to eleven times as fast. Eight platform businesses outside Indonesia show the same boundary with smaller year-to-year divergence. The proposal compares these records rather than treating any one of them as the measure of digital activity.

1. Introduction

Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). That figure is one record of the activity: the gross value of the transactions involved. The platforms that carry those transactions report a different number, their own revenue; national statistical agencies report a third, the online sales of surveyed businesses; and payment systems report a fourth, the value of funds moved. Each is correct for what it measures. These measures are often used interchangeably even though they do not measure the same thing, and this paper argues that doing so produces systematically wrong conclusions about the scale and growth of platform economies.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as revenue. The remaining 90 are still recorded within the platform's system but do not constitute platform revenue; they represent merchant receipts, driver payouts, inventory costs, taxes, and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge. Neither number is wrong. Revenue records what the platform earned and transaction value records what it carried; questions about the size and growth of the digital economy concern what was carried, while the figure a listed platform must report in its financial statements is what it earned.

Indonesia, the largest digital market in Southeast Asia, shows how far apart these records are. The study covers the five Indonesian platforms that publish usable transaction and revenue figures: the marketplaces Tokopedia, Shopee, Blibli and Bukalapak, and Grab, whose Indonesian business is mainly delivery and other on-demand services. In 2023, Tokopedia, Shopee and Grab together processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion. BPS-Statistics Indonesia, the national statistical agency, placed the whole marketplace channel at between US\$13.2 billion and US\$23.7 billion, depending on how sales through several channels are attributed, and all e-commerce at US\$72.3 billion. These figures cover different scopes, and that is the point: each answers a different question, and anyone quoting one of them as the size of the digital economy has chosen a record.

The records also disagree about growth. Between 2022 and 2023, Tokopedia's revenue rose 53.2 percent while the transaction value it processed fell 8.9 percent: read through its accounts the platform was expanding quickly, read through its transactions it was shrinking. The reversal is not

isolated. Among the three platforms that report both figures themselves, revenue grew faster than transaction value in six of eight year-to-year comparisons between 2020 and 2025, and the two moved in opposite directions twice. Nationally, BPS reports e-commerce value up 17.1 percent between 2023 and 2024, with 98.5 percent of the increase outside the marketplace channel that platform accounts describe, while payment measures from Bank Indonesia, the central bank, grew between 1.8 and 11 times as fast.

A statement about how large Indonesia's digital economy is, or how fast it grew, is therefore a statement about a record. An investor reading platform revenue, an analyst sizing e-commerce from marketplace figures and a regulator building a rule on marketplace records will each reach a different conclusion about the same years. Indonesia's Minister of Finance Regulation No. 37 of 2025 (PMK 37/2025) is one such rule, and Section 5.3 shows that it rests on the channel that carried 1.5 percent of recent e-commerce growth.

1.1 Research question and objectives

One question drives the proposal:

How much economic activity remains invisible when digital platforms are measured through their reported revenue?

Indonesia provides the main empirical setting. Three objectives follow: first, to measure the invisible wedge across the platform histories available from 2020 to 2025; second, to explain large movements using disclosed changes in monetization, customer incentives, revenue recognition, and reporting scope; and third, to compare the platform evidence with national e-commerce and payment records, to determine whether they give the same account of economic scale and growth and what their disagreement implies for measures built on a single record.

A separate comparison of eight listed platform businesses outside Indonesia, including eBay, Etsy, Shopify and Mercado Libre, tests whether the same separation between transaction value and revenue appears under other business models.

1.2 Contributions

First, the study turns the invisible wedge from a one-year estimate into a measure followed over time, and shows that revenue is not a stable guide to the commerce behind it: among the issuer-reported platforms, revenue outgrew transaction value in six of eight comparisons and moved against it twice.

Second, I explain why the wedge moves using accounting components the platforms disclose. Of the increase in Tokopedia's net revenue between 2022 and 2023, 60.6 percent reconciles to lower customer incentives and 39.4 percent to higher gross revenue, in a year when its transaction value fell.

Third, I extend the comparison beyond company accounts and show that national statistics and payment records disagree with the platform picture, and with each other, about both the level and the growth of Indonesian digital commerce.

Fourth, the study separates the figures issuers report from those that must be constructed, and shows that the longitudinal result rests on the reported figures alone.

1.3 Positioning the contribution

The contribution is a finance measurement problem before it is a tax-policy one. Platform accounts make the intermediation boundary observable, but that boundary affects the conclusions investors, researchers, statistical agencies, and policymakers draw about scale, growth, and monetization. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter, while Berg et al. (2020) show that intermediary-held digital traces contain information conventional records can miss. This paper applies those ideas to platform economies and shows that the record chosen to describe digital activity changes the economic conclusion.

2. Literature Review and Research Gap

Digital platforms sit between economic activity and the records used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical, and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking how much activity remains invisible when platforms are measured through reported revenue.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

Accounting can widen or narrow this difference. Under IFRS 15, whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. The literature therefore explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, the quantities are published by different producers. Industry reports size the regional digital economy by gross merchandise value (Google, Temasek and Bain 2024), national statistics measure the online sales of surveyed businesses (BPS-Statistics Indonesia 2025), and listed platforms report the revenue they recognize under IFRS 15 (International Accounting Standards Board 2014). A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and growth. This thesis therefore compares the records rather than assuming that one can substitute for another.

2.3 Digital Records and Third-Party Information

Activity outside platform revenue is still recorded. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records while their transactions leave detailed records inside a digital platform. This differs from the usual shadow-economy problem, where part of the difficulty is that economic activity leaves few reliable formal records (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Digital records can contain useful economic information that conventional records miss (Berg et al. 2020), and platform-mediated activity can create additional financial and administrative traces (Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

These records also matter once they become available to government. Public-finance research shows that compliance and enforcement change when information about income or transactions is independently reported by third parties (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules (OECD 2020) and the European Union's DAC7 regime (European Union 2021) apply this principle to digital platforms, and PMK 37/2025 follows the same logic through seller identification, transaction-linked reporting, and marketplace withholding. The relevance here is informational: records outside platform revenue exist, and what they can show depends on which of them a measure or a rule draws on.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; official measurement frameworks explain why these quantities should not be treated as equivalent; and research on digital records explains why activity outside revenue is still observable.

What remains less clear is how large these differences become in practice, whether they persist over time, and whether choosing one record instead of another changes the economic conclusion. This thesis addresses that gap by measuring the invisible wedge over time, examining disclosed reasons for major movements, comparing platform accounts with national e-commerce and payment records, and using external platform evidence to test whether the same pattern appears beyond Indonesia.

3. Theoretical Framework and Hypotheses

Digital platform activity can be observed through several records, but those records do not measure the same economic object. The framework begins with the difference between the value a platform processes and the revenue it recognizes, then asks whether that relationship remains stable over time and whether other records produce the same account of digital activity.

For the platform analysis, transaction value and revenue are paired only when they refer to the same period and business activity. Figures that issuers report and figures that must be constructed are kept apart, as are national statistics and payment records (Section 4).

3.1 Measurement Framework

Let V_{it} denote transaction value processed by platform i in period t , and R_{it} the revenue recognized over the same matched period and business scope. The absolute invisible wedge is:

$$W_{it} = V_{it} - R_{it}$$

W is transaction value outside the platform's recognized revenue; it is not merchant profit, worker income, taxable income, unpaid tax, tax evasion, or value added. Because platforms differ in size, I also express the wedge relative to revenue. The Ecosystem Ratio is:

$$E_{it} = (V_{it} - R_{it}) / R_{it}$$

An E of 9 means that for every one unit of platform revenue, nine units of transaction value lie outside the revenue line. Seen from the other side, E is the platform's monetization rate, $t = R/V$, inverted: $E = 1/t - 1$, so a platform keeping 2.5 percent of transaction value has an E of 39. It is reported this way because the question concerns the commerce outside revenue rather than the share the platform captures.

The Ecosystem Ratio is constructed for this study rather than adopted from prior work. Its basis is the platform-economics result that booked revenue need not track the transaction value a multisided platform coordinates (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016), together with the accounting result that the share of transaction value entering revenue is itself a reporting choice (International Accounting Standards Board 2014; De Franco, Kothari and Verdi 2011).

A single ratio does not show whether the relationship holds through time, so I define annual growth divergence as:

$$D_{it} = g(R_{it}) - g(V_{it})$$

A positive D means revenue grew faster than transaction value, so the platform captured a larger share and E fell; a negative D means the reverse. If revenue were a stable proxy for transaction value, D would stay close to zero. Together, W, E, and D describe the size of the wedge and how it changes over time.

3.2 Hypotheses

The four hypotheses test the research question one record at a time. H1 concerns transaction value and revenue within platforms, H2 and H3 the national e-commerce evidence, and H4 the comparison across records. None is interpreted as a causal claim. Table 1 pairs each hypothesis with the evidence that tests it.

H1. Within the same platform, transaction value and platform revenue do not grow in proportion.

If they did, D would stay close to zero and E would be constant. The test is whether D departs from zero and changes sign.

H2. National e-commerce growth comes mainly from more participating businesses rather than from higher sales per business.

Growth driven by new sellers can take place in channels that platform accounts do not cover.

H3. Marketplace participation is associated with stronger financial recordkeeping.

Recordkeeping determines whether activity outside platform accounts leaves a record that statistics can observe. The relationship is tested as an association.

H4. Platform accounts, national e-commerce statistics and payment records give different accounts of the level and growth of Indonesian digital commerce.

H4 is the central claim of the thesis; H1 to H3 supply its components.

Table 1. Hypotheses and principal empirical tests.

Hypothesis	Principal empirical test
H1 - Transaction-revenue divergence	Growth divergence D and the Ecosystem Ratio for the platforms that report both figures
H2 - Sources of e-commerce growth	BPS national e-commerce value split into business count and sales per business
H3 - Marketplace participation and recordkeeping	BPS published comparison of marketplace and non-marketplace sellers
H4 - Measurement choice	Levels and growth rates reported by platform accounts, BPS and Bank Indonesia for the same years

The comparison of platform businesses outside Indonesia corroborates H1 rather than forming a fifth hypothesis.

4. Data and Methodology

The study uses several records because no single one captures every part of digital platform activity. Platform reports provide transaction value and revenue; BPS-Statistics Indonesia provides national e-commerce and business evidence; Bank Indonesia provides payment statistics; and platform companies outside Indonesia provide a separate comparison. The records answer different questions and are not combined into one dataset.

4.1 Selecting the Platform Evidence

The basic observation is one platform in one year, with transaction value and revenue referring to the same period and, as closely as possible, the same business activity. An observation is retained only when its period, geographic and business coverage, units, definitions, and source can be identified; measures referring to different periods are excluded, and major changes in accounting definitions or business structure are recorded.

The sample consists of the Indonesian platforms that publish both transaction value and revenue. Lazada and TikTok Shop, both large in Indonesia, are outside it because no matched Indonesian revenue series has been located for either. The window runs from the first matched pair, Bukalapak in 2020, to the latest, Blibli in 2025.

Table 2. Indonesian platform evidence.

Type of evidence	Platforms	Annual observations	Role
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	2	Main sample: cleanest scope, shortest window
Issuer-reported, broader scope	Blibli third-party business; Bukalapak Group	9	Main sample: most of the time series
Requires construction	Grab; Shopee Indonesia	6	Supporting: 2023 scale comparison

Table 2 separates the pairs issuers report themselves from the country figures that must be constructed. The main sample is the first group: 11 platform-years yielding 8 year-to-year comparisons. Adding Grab and Shopee gives 17 platform-years and 12 comparisons.

Each series covers the years in which a matched pair exists. Tokopedia is an Indonesia-aligned segment rather than a literal country line and enters only for 2022 and 2023: the 2021 pair covers mismatched periods, and the Tokopedia and TikTok Shop businesses were combined under PT Tokopedia on 31 January 2024, after which the listed parent reports a contractual fee rather than platform revenue. Blibli's reported pairs run from 2021 to 2025, and its third-party segment includes online travel. Bukalapak reports at Group level, including some overseas activity, from 2020 to 2023; its 2024 pair is excluded because transaction value covers nine months and revenue

twelve. Grab is observed from 2021 to 2023, after which it reports transaction value on a different basis, and Shopee from 2022 to 2024.

4.2 Comparing Platform Activity Through Time

Changes through time are examined mainly by comparing each platform with itself. This keeps the company, business activity, and reporting convention as similar as the disclosures allow. When transaction value and revenue move sharply apart, disclosed changes in monetization, customer incentives, revenue recognition, and business scope are used to reconcile the movement where the filings permit; these reconciliations are arithmetic rather than causal.

Companies use different names for transaction activity, including gross transaction value (GTV), gross merchandise value (GMV), and total payment volume (TPV). Transaction value, V, is the umbrella concept in this proposal and GTV is the label used for it, while each issuer's own label and definition are retained and checked rather than assumed to be identical.¹

Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D. US dollar values are used only for the 2023 cross-section, where rupiah figures are converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar (World Bank n.d.).

4.3 Direct and Constructed Indonesia Measures

Not every company reports the same Indonesia-specific information. Table 3 sets out, for each platform, what is disclosed directly, what has to be constructed, and the limitation that follows.

Table 3. Construction of the principal platform measures.

Platform	Directly available	Constructed or external input	Main limitation
Tokopedia	E-commerce transaction value and segment revenue	None for matched pair	Indonesia-aligned segment, not literal country line
Grab	Indonesia revenue; Group revenue and GTV	Indonesia GTV derived from Group figures	Indonesian ratio equals the Group ratio
Shopee	Group service monetization rate	External Indonesia transaction estimate; Indonesia revenue derived	Indonesian ratio set by the Group rate
Blibli / Bukalapak	Reported transaction and revenue pairs	None for retained pairs	Scope extends beyond Indonesian goods commerce

Grab reports Indonesian revenue directly but not Indonesian transaction value. Indonesian GTV is therefore estimated as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The Group monetization rate is platform revenue divided by gross transaction value.

Sea Limited does not report Shopee transaction value separately for Indonesia. Indonesian transaction value is taken from Momentum Works, a Singapore-based industry research firm whose annual Southeast Asian e-commerce report is widely cited for country-level platform transaction estimates. It is used because no issuer discloses Shopee's Indonesian transaction value, it is labeled as a third-party estimate wherever it appears, and Indonesian revenue is then estimated using Sea's Group service monetization rate:

¹ Issuer labels differ. GoTo reports GTV and Sea Limited reports GMV for Shopee; for these the label differs but the measure is the same. Blibli and Bukalapak report TPV for the segments used here, and Grab reports GMV for deliveries and mobility and TPV for financial services. Where a source reports TPV the measure rests on a different basis, so it is retained under its own label rather than relabeled as GTV.

$$\text{Indonesia revenue} = \text{Indonesia transaction value} \times \text{Group monetization rate}$$

In both cases the Indonesian ratio is fixed by construction: Grab's Indonesian E equals its Group E, and Shopee's is the inverse of Sea's monetization rate less one. The two series therefore contribute Indonesian levels, through Grab's reported revenue and the external Shopee estimate, but no independent Indonesian evidence on the ratio or its movement, which is why they are outside the main sample.

4.4 Evidence Beyond Platform Accounts

Table 4. Records used and their role.

Evidence source	What is observed	Role
Indonesian platform histories	Transaction value and revenue through time	Measure the wedge and its movement
BPS national e-commerce statistics	Online sales of surveyed businesses, by channel, and the number of businesses	Examine the level and sources of growth
BPS marketplace study	Business-level marketplace and recordkeeping evidence	Examine the association with financial recordkeeping
Bank Indonesia	Economy-wide digital payment measures	Compare payment growth with e-commerce growth
Platform companies outside Indonesia	Matched transaction and revenue measures	Test whether the same boundary appears elsewhere

Table 4 gives each record a single role. The records answer different questions, so they are reported separately rather than pooled into one estimate.

For H2, national e-commerce value S is written as:

$$S = N \times A$$

where N is the estimated number of e-commerce businesses and A is implied nominal sales per business, and the change in S is divided arithmetically between the two. BPS publishes two counts of 2023 businesses; the count consistent with its stated growth figure is used, and the other is retained as a sensitivity check.

For H3, BPS's published business-level comparison of marketplace and non-marketplace sellers is used. For H4, Bank Indonesia payment measures are compared with e-commerce growth on growth rates only: they cover payments across the whole economy rather than e-commerce alone, so they are never added to platform or BPS values.

4.5 External Evidence and Robustness

The external comparison contains 48 matched annual observations for eight listed platform businesses from 2017 to 2025: eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify and Zalando. It tests whether the same separation between transaction value and revenue appears under other business models and is reported separately from the Indonesian evidence.

Robustness checks vary the admission rules, revenue definitions, publication versions, and the assumptions in the constructed country measures. The resulting ranges are reported as sensitivity ranges rather than statistical confidence intervals.

5. Preliminary Evidence and Feasibility

The evidence assembled so far establishes that the proposed comparisons can be carried out and that the choice of record produces economically meaningful differences. The results first show the scale and movement of the wedge, then compare the platform picture with national and payment records.

5.1 Platform Scale and Movement

The 2023 comparison covers Shopee and Tokopedia, which together represented about 70 percent of estimated Indonesian marketplace transaction value, and Grab, a major delivery intermediary.

Table 5. Indonesia-focused 2023 platform comparison, US\$ billions.

Case	Transaction value V	Revenue R	Wedge W	Ecosystem Ratio E	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Transaction value constructed
Tokopedia	16.331	0.405	15.926	39.296x	Issuer-reported pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Transaction value and revenue estimated
Three platforms	43.233	3.162	40.070	12.671x	Sum of three cases, not national total

Table 5 reports that cross-section. Across the three platforms, US\$43.23 billion of transaction value corresponds to US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion and a ratio of 13.7 to 1. The wedge is equivalent in magnitude to about 2.9 percent of Indonesia's 2023 nominal GDP of US\$1.371 trillion (World Bank n.d.); the comparison is of magnitude only, since the wedge is transaction value rather than value added.

Two of the three ratios rest on construction. Shopee's 9.0 is the inverse of the assumed Group monetization rate less one, and Grab's 7.9 is its Group ratio. Varying Shopee's rate between 9 and 11 percent moves the combined wedge only between US\$39.86 billion and US\$40.29 billion, and excluding any one platform leaves at least US\$20.70 billion.

Blibli's third-party business and Bukalapak Group, which report both figures themselves, carried wedges of US\$3.20 billion and US\$10.50 billion in 2023 at the same exchange rate. They are not added to the total because their measures include online travel and Group-level activity, but their ratios, 43.4 and 36.0, sit close to Tokopedia's 39.3: the three issuer-reported ratios lie between 36 and 43, while the two constructed ratios reflect Group monetization.

The relationship is not stable. Across the eight year-to-year comparisons in the main sample, revenue grew faster than transaction value in six, with a median absolute gap of 38.92 percentage points between the two growth rates, and the two moved in opposite directions twice. Without the largest comparison, revenue still grew faster in five of seven, with a median gap of 15.74 points, and excluding any single platform leaves revenue faster in most comparisons and at least one reversal. Including Grab and Shopee gives ten of twelve, but those four comparisons track Group monetization rather than Indonesian activity. Figure 1 shows the Ecosystem Ratio for each platform; every series ends lower than it began, though not by a steady decline.

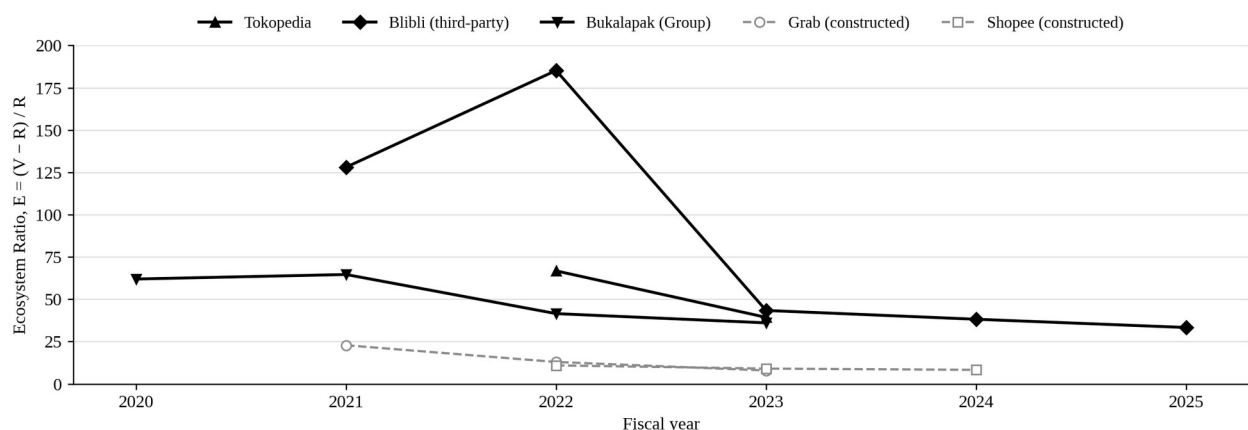


Figure 1. Ecosystem Ratio by platform, 2020–2025. Solid lines are issuer-reported pairs; dashed lines are constructed Indonesian series whose ratios equal Group ratios by construction.

Tokopedia shows why the two measures separate. Between 2022 and 2023 its transaction value fell 8.9 percent while third-party net revenue rose 53.2 percent; 60.6 percent of the revenue increase reconciles to lower customer incentives and 39.4 percent to higher gross revenue. Read through its accounts the platform was growing quickly, read through its transactions it was shrinking. Blibli shows the same reversal in 2024–2025, when transaction value fell 1.89 percent and net revenue rose 12.07 percent, so the pattern is not confined to one platform or one year.

5.2 Evidence Beyond Platform Revenue

National statistics give a different account of the same commerce, beginning with its level. BPS measures the online sales reported by surveyed e-commerce businesses, whereas platform transaction value is the gross value of orders processed, and the external Shopee estimate explicitly includes cancelled and returned orders. For 2023, BPS places the marketplace channel at Rp200.68 trillion, about US\$13.2 billion, in its exclusive split of transaction value by sales channel, and at 32.74 percent of the total, about US\$23.7 billion, in its multiple-response measure, which counts businesses under every channel they use. Tokopedia and Shopee alone reported US\$37.9 billion. Both records are correct for what they measure; reconciling their definitions is part of the thesis.

Growth tells the same story. Table 6 summarizes the evidence corresponding to H2–H4.

Table 6. Preliminary evidence beyond platform accounts.

Question	Preliminary evidence	Interpretation
Where does e-commerce growth come from?	E-commerce value +17.08%; estimated business count +15.31%; sales per business +1.54%. The business count accounts for 90.29% of the increase (70.95% under BPS's other count)	Growth comes mainly from more businesses selling online (H2)
Where does that growth occur?	Marketplace channel +1.45%; other channels +20.57%; 98.46% of the increase is outside the marketplace channel	Marketplace records capture little of recent growth (H4)
Are marketplace use and financial records related?	Complete financial statements: 28.63% of marketplace users versus 12.25% of non-marketplace users	Marketplace participation is associated with stronger recordkeeping (H3)
Do payment records show the same growth?	BPS e-commerce +17.08%; electronic-money shopping +30.47%; mobile-banking payments and purchases +82.84%; QRIS +186.98%	Economy-wide payment measures grow 1.8 to 11 times as fast (H4)

Outside the marketplace channel, e-commerce runs mainly through social media and instant messaging. In 2023, 95.33 percent of e-commerce businesses sold through instant messaging, and social media accounted for 44.04 percent of attributed transaction value against 32.74 percent for marketplaces. Growth in those channels is growth that the marketplace platforms studied here do not record. The business-count split is arithmetic, the recordkeeping result is an association, and payment values are compared on growth only.

5.3 Feasibility and Scope of the Preliminary Findings

The preliminary evidence supports each hypothesis and shows that the comparisons can be completed. It supports H1 through the six of eight comparisons in which revenue outgrew transaction value; H2 through the business-count share of growth; H3 through BPS's business-level comparison; and H4 through the disagreement in both level and growth between platform, national and payment records.

The external comparison corroborates the direction of H1 and qualifies its size. Transaction value exceeds revenue in all eight businesses, with monetization rates from about 2.4 percent at Shopify to about 25 percent at Mercado Libre among the marketplaces. Among the 29 comparisons free of acquisitions or perimeter changes, revenue grew faster than transaction value in 22 and the two moved in opposite directions in 4, but the median gap, 7.1 percentage points, is far smaller than in the Indonesian main sample.

PMK 37/2025 shows why the choice of record matters beyond statistics. Administered by the Directorate General of Taxes (DJP), it requires designated marketplace operators to report seller-linked transactions and withhold tax, with implementation from 1 November 2026. Because it is built on the marketplace record, its reach is bounded by the marketplace channel, which carried 15.8 percent of 2024 e-commerce value and 1.5 percent of 2023–2024 growth. The regulation is examined as a rule built on one record; its compliance and revenue effects lie outside this study.

At this stage, the proposed measurement can be constructed, followed through time, reconciled to disclosed accounting changes, and compared with independent records of Indonesian digital commerce.

6. Why the Problem Matters

These records are designed to measure different things, and the problem arises when one is used as a substitute for another. Platform revenue measures what the platform earns, transaction value the commerce it carries, national e-commerce statistics the online sales of businesses, and payment systems the movement of funds. Between 2022 and 2024 they disagreed about both the size and the growth of Indonesian digital commerce: Tokopedia's transactions shrank while its revenue grew, the marketplace channel was almost flat while national e-commerce rose 17 percent, and payment measures rose by as much as 187 percent. A reader who takes any one of them as the measure of the digital economy, whether to value a platform or to size the sector, inherits that record's view.

The same holds for rules: a reporting regime built on the marketplace record reaches only what that record covers.

7. Limitations and Robustness

The evidence is descriptive, and its limits are specific. The main sample is small: three issuer-reported platforms, 11 platform-years and 8 comparisons, which supports description rather than statistical inference. Its scope is uneven: Tokopedia is an Indonesia-aligned segment observed for two years, Blibli's segment includes online travel, and Bukalapak reports at Group level. Grab's and Shopee's Indonesian figures are constructed, so their ratios carry no independent Indonesian information, and Shopee's revenue rests on an assumed monetization rate. The records compared in Section 5.2 are defined differently, so their disagreement is measured at this stage rather than reconciled.

Table 7 pairs each supported conclusion with what it does not establish.

Table 7. Main inference boundaries.

Evidence supports	Does not by itself establish
W, E, and D describe the size and movement of the wedge.	Participant profit, taxable income, unpaid tax, tax evasion, or missing GDP.
Issuer-reported pairs show revenue and transaction value moving apart within platforms.	Independent Indonesian ratio evidence from the constructed Grab and Shopee series.
Disclosed revenue components reconcile selected movements in the wedge.	A causal effect of incentives, monetization, or accounting choices.
BPS evidence describes the sources and channels of e-commerce growth and recordkeeping associations.	Causal business entry, productivity effects, or marketplace-induced formalization.
Platform, BPS and payment records disagree in level and growth.	That either record is wrong, or that payment values measure sales.
PMK 37/2025 illustrates a rule built on the marketplace record.	A completed compliance or tax-revenue effect.

Robustness is reported as sensitivity ranges: Shopee's assumed rate is varied, each platform is excluded in turn, the largest comparison is dropped, and the longitudinal result is restricted to issuer-reported pairs.

8. Next Steps

The data and empirical framework are assembled. The remaining work is to confirm with the committee the treatment of the broader-scope and constructed series, reconcile the definitions behind the BPS and platform levels, divide movements in the wedge into changes in transaction volume and in monetization, and complete the robustness checks. Table 8 sets out that schedule.

Table 8. Planned work toward the final thesis.

Period	Planned work
Proposal stage	Incorporate examination feedback; confirm the main sample and the treatment of broader-scope and constructed series.
October–November 2026	Reconcile BPS and platform definitions; divide wedge movements into volume and monetization effects; complete robustness checks; finalize tables and figures.
Before the final defense	Complete the full manuscript and the final citation and consistency review.

The proposed study begins from a simple distinction between the value a platform processes and the revenue it recognizes. The preliminary evidence shows that the records of Indonesia's digital commerce disagree about its size and its growth, and that the disagreement is large enough to change the conclusion a reader draws. The thesis measures that disagreement and identifies which conclusions depend on the record chosen.

Appendix A - Variables and Data Sources

Table A1. Variable definitions and sources.

Variable	Definition	Source
V	Transaction value processed through the platform in the period	Issuer disclosure, or construction (Table 3)
R	Revenue the platform recognizes for the same period and business scope	Issuer disclosure, or construction (Table 3)
$t = R / V$	Monetization rate: the share of transaction value recognized as revenue	Computed
$W = V - R$	Invisible wedge: transaction value outside recognized revenue	Computed
$E = (V - R) / R = 1/t - 1$	Ecosystem Ratio: wedge per unit of revenue; constructed for this study	Computed
$D = g(R) - g(V)$	Growth divergence: revenue growth minus transaction-value growth	Computed from consecutive years
$S = N \times A$	National e-commerce value: number of businesses times sales per business	BPS-Statistics Indonesia

Table A1 defines each variable and says where it comes from. Table A2 traces each record to the filings and publications behind it.

Table A2. Source lineage by evidence type.

Evidence type	Platform / source	Transaction-value source	Revenue / comparison source
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Issuer-reported, broader scope	Blibli third-party business	Global Digital Niaga prospectus and results releases	Same issuer filings
Issuer-reported, broader scope	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Requires construction	Grab Indonesia	Derived from Indonesia revenue and Group monetization rate	Grab Form 20-F
Requires construction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023 and 2024; marketplace study	Business-count, channel and recordkeeping evidence
Payment statistics	Bank Indonesia	Payment System Statistics and QRIS reports	Growth-rate comparison only
Exchange rate and GDP	World Bank	World Development Indicators, 2023	Converts 2023 rupiah values; GDP scale comparison
External corroboration	eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify, Zalando	Issuer transaction measures	Issuer revenue measures
Regulatory	PMK 37/2025; DJP	Ministry of Finance / Directorate General of Taxes	Institutional architecture only

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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